

Telecom Diode Laser Reference Manual

Waveguide Mode Physics & Drift-Diffusion Formulations

1. Executive Summary

This reference manual documents the complete physical formulations and parameters of our multiscale telecom laser simulator. By integrating 2D transverse carrier transport with a unified wave-optical solver, we capture the core physics of ridge-waveguide lasers. The independent PhotonSolver is retired; photon distribution and local stimulated emission are computed directly from the mode intensity profile, ensuring full physical consistency.

2. Electrostatic & Carrier Transport

Carrier transport in the 2D transverse plane (y,z) is modeled via the Poisson equation coupled with the electron and hole drift-diffusion equations. The electrostatic potential is determined by:

$$-\nabla \cdot (\epsilon_0 \epsilon_r \nabla \psi) = q(p - n + N_D - N_A)$$

The electron and hole current densities J_n and J_p contain both drift (driven by electric fields) and diffusion (driven by carrier density gradients):

$$\mathbf{J}_n = -q\mu_n n \nabla \psi + qD_n \nabla n$$

$$\mathbf{J}_p = -q\mu_p p \nabla \psi - qD_p \nabla p$$

The steady-state conservation equations couple current divergence directly with recombination sinks:

$$\nabla \cdot \mathbf{J}_n = qR_{\text{total}}, \quad -\nabla \cdot \mathbf{J}_p = qR_{\text{total}}$$

3. Unified Wave-Optical Solver & Photon Representation

In this simulator, we retired the dedicated PhotonSolver. The local photon density $s_{\text{ph}}(y,z)$ is represented directly through the mode intensity computed from the transverse wave equations, scaled by the average cavity power P_{cav} solved by the 1D longitudinal traveling wave equations:

$$s_{\text{ph}}(y, z) = \frac{\bar{P}_{\text{cav}}}{h\nu \cdot v_g} \frac{|\mathbf{E}_{2\text{D}}(y, z)|^2}{\iint |\mathbf{E}_{2\text{D}}|^2 dy dz}$$

The local stimulated emission rate R_{stim} is formulated directly using this optical mode overlap:

$$R_{\text{stim}}(y,z) = v_g * g(y,z,\omega) * s_{\text{ph}}(y,z)$$

where v_g is the group velocity and g is the local material gain. This guarantees that spatial hole burning (stimulated carrier depletion) occurs only where the optical field is present.

4. Complex Transverse Helmholtz Equation

The transverse waveguide modes are solved via the complex 2D vector wave equation. Assuming TE polarization, this separates into a complex scalar eigenvalue problem:

$$\nabla_t^2 \mathbf{E}_{2\text{D}} + k_0^2 \tilde{\epsilon} \mathbf{E}_{2\text{D}} = \tilde{\beta}^2 \mathbf{E}_{2\text{D}}$$

Here, $\lambda = \beta_{\text{tilde}}^2$ is the complex eigenvalue. The operator is non-Hermitian due to the complex permittivity $\epsilon_{\text{tilde}}(y,z,\omega)$ representing active amplification and facet escape losses:

$$\tilde{\epsilon}(y, z, \omega) = n_r^2 - i \frac{n_{rc}}{\omega} [g(y, z, \omega) - \alpha_i(y, z) - \alpha_{\text{escape}}]$$

5. Empirical Refractive Index & Gain Models

The material gain spectrum $g(y,z,\omega)$ inside the active MQW region is modeled using an empirical parabolic decay around the peak transition energy E_{peak} :

$$g(y, z, \omega) = F_{\text{QW}}(y, z) [g_{\text{peak}}(n, p, T) - a_{\omega}(\hbar\omega - E_{\text{peak}})^2]$$

The peak gain g_{peak} depends logarithmically on the carrier density product and exponentially on the temperature (to represent thermal carrier spillover):

$$g_{\text{peak}}(n, p, T) = g_0 \ln \left(\frac{np + \varepsilon}{n_{\text{tr}}p_{\text{tr}} + \varepsilon} \right) e^{-\frac{T - T_0}{T_{\text{char}}}}$$

The real index $n_r(y,z,\omega)$ includes Adachi bulk dispersion, plus temperature (thermo-optic) and carrier (plasma dispersion) perturbations:

$$n_r(y, z, \omega) = n_{\text{bulk}}(\omega) + \frac{dn_r}{dT}(T - T_0) + \frac{dn_r}{dn}n$$

The bulk refractive index dispersion of the semiconductor layers is modeled using the Adachi formulation:

$$n_{\text{bulk}}^2(\omega) = A + B \frac{E_{\text{ph}}^2}{E_{\text{ph}}^2 - (\hbar\omega)^2}$$

6. Lattice Heat Transport (ThermoSolver)

ThermoSolver calculates the 2D spatial temperature distribution $T(y,z)$ within the laser package. The steady-state lattice heat flow is governed by the heat conduction equation:

$$-\nabla \cdot (k_L \nabla T) = Q_{\text{gen}}$$

Where k_L is the heterogeneous thermal conductivity of the layers, and Q_{gen} is the total heat generation density (W/cm³). The heat source term combines Joule heating (free-carrier transport), non-radiative recombination heating, and optical absorption heating:

$$Q_{\text{gen}} = \mathbf{J} \cdot \mathbf{E} + E_g R_{\text{SRH}} + E_g R_{\text{Auger}} + \hbar \omega v_g \alpha_{\text{abs}} S_{\text{ph}}$$

Where:

- $\mathbf{J} \cdot \mathbf{E} = -(\mathbf{J}_n + \mathbf{J}_p) \cdot \text{grad}(\psi)$ is the Joule heating term.
- $E_g \cdot R_{\text{SRH}}$ and $E_g \cdot R_{\text{Auger}}$ represent thermal energy released during non-radiative carrier recombination.
- $\hbar \omega v_g \alpha_{\text{abs}} S_{\text{ph}}$ represents the optical absorption heating.

Cooling is modeled via convective package boundaries at the copper submount interface:

$$-k_L \frac{dT}{dn} = h_{\text{conv}} \cdot (T - T_{\text{sub}})$$

where h_{conv} is the heat transfer coefficient (e.g. 1.5 W/cm²-K) and T_{sub} is the submount sink temperature (e.g. 293 K).

7. Material Parameter Tables (InP / InGaAsP System)

The table below details typical parameter values and ranges used in 1.31 μm and 1.55 μm telecom laser simulators, compiled from standard literature.

Parameter Name	InP Cladding	InGaAsP SCH/QW	Unit	Reference
Refractive Index (n_{bulk})	3.17 - 3.21	3.38 - 3.52	-	Piprek, Opto. Devices
Bandgap Energy (E_g)	1.35	0.75 - 0.95	eV	Vurgaftman et al. (2001)
Electron Mobility (μ_n)	3000 - 4500	2000 - 3500	$\text{cm}^2/\text{V-s}$	Coldren & Corzine
Hole Mobility (μ_p)	100 - 150	70 - 120	$\text{cm}^2/\text{V-s}$	Coldren & Corzine
SRH Recomb. Lifetime (τ)	2.0 - 5.0	1.0 - 3.0	ns	Piprek, Semiconductor Lasers
Radiative Coeff. (B_{sp})	$1.0\text{e-}10$	$1.5\text{e-}10$ - $2.5\text{e-}10$	cm^3/s	Vurgaftman et al. (2001)
Auger Coefficient (C)	$1.0\text{e-}30$	$1.0\text{e-}29$ - $5.0\text{e-}29$	cm^6/s	Coldren & Corzine
Thermo-Optic (dn/dT)	$2.0\text{e-}4$	$2.5\text{e-}4$	K-1	Piprek, Opto. Devices
Plasma Dispersion (dn/dn)	$-1.8\text{e-}20$	$-2.5\text{e-}20$	cm^3	Coldren & Corzine
Thermal Cond. (k_T)	0.68	0.05 - 0.12	W/cm-K	Piprek, Semiconductor Lasers

8. Coupled Solver Numerical Workflow

The self-consistent loop follows a structured sequence at each iteration:

1. CarriersSolver: Computes carrier densities (n, p) and potential (ψ) with scaled Jacobian.
2. Complex Permittivity Update: Updates real index n_r and active region gain g .
3. Helmholtz Solve (EMSolver): Solves wave modes to output mode profiles E_{2d} and eigenvalues.
4. 1D Longitudinal Feedback: Calibration loop updates escape loss rate α_{escape} .
5. ThermoSolver: Calculates package heat generation and thermal profiling.

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